

From Droop-Only to Advanced Control: Progressive Design of a Standalone Grid-Forming Inverter

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Abstract—The increasing penetration of inverter-based renewable energy sources has significantly reduced the inertia of modern power systems, motivating the adoption of grid-forming (GFM) inverters for stable islanded operation. Among various GFM control strategies, droop control is widely used due to its simplicity and decentralized nature. However, droop-only control may exhibit limited damping and degraded dynamic performance under load disturbances, even for a single inverter. This paper presents a progressive control design for a single grid-forming inverter, starting from conventional droop-only control and systematically incorporating advanced control layers to improve dynamic stability and voltage–frequency regulation. A detailed system model is developed in the synchronous reference frame, and the impact of load variations, particularly inductive loads, is analyzed. Simulation results demonstrate that the proposed progressive control design significantly mitigates frequency oscillations, enhances damping, and improves transient performance compared to droop-only operation.

Index Terms—Grid-forming inverter, Frequency oscillation, Droop control, Damping improvement, Inductive load, Dynamic performance, Inner control loops, Virtual impedance.

I. INTRODUCTION

The rapid integration of inverter-based renewable energy resources has fundamentally altered the dynamic characteristics of modern power systems. As conventional synchronous generators are progressively displaced, the overall system inertia is significantly reduced, leading to increased challenges in frequency and voltage regulation, particularly under islanded and weak-grid operating conditions [6]. In this context, grid-forming (GFM) inverters have emerged as a key enabling technology due to their ability to establish voltage and frequency references independently of the grid [9].

Among the various control strategies proposed for GFM inverters, droop control remains one of the most widely adopted approaches owing to its simple structure, decentralized implementation, and intuitive power–frequency and voltage–reactive power regulation characteristics [8]. Droop-based GFM control enables autonomous operation without reliance on phase-locked loops or communication infrastructure, making it particularly suitable for islanded microgrids and low-inertia power systems. However, despite these advantages, droop-only control exhibits inherent limitations in dynamic performance.

In particular, the lack of explicit damping mechanisms in droop-only control may result in lightly damped or oscillatory

responses during load disturbances, especially in systems dominated by inductive loads. Under such conditions, frequency and voltage deviations can persist following sudden changes in load demand, and system performance becomes highly sensitive to the selection of droop coefficients.

To overcome these limitations, recent studies have investigated the integration of additional control layers, such as inner voltage and current control loops, virtual impedance, and damping enhancement techniques, to complement conventional droop control [4]. These advanced control structures aim to improve transient response, enhance system damping, and ensure robust operation across a wide range of loading conditions. Nevertheless, existing studies often present advanced grid-forming control methods as isolated architectures without systematically illustrating how successive control layers modify system dynamics and stability characteristics. A structured design-oriented investigation that progressively connects droop-only operation with advanced inner-loop–based control for a single grid-forming inverter remains relatively limited.

This paper presents a progressive and design-oriented control framework for a single grid-forming inverter, beginning with conventional droop-only control and systematically extending toward advanced control structures. Rather than proposing a completely new control algorithm, the work emphasizes how each additional control layer influences damping behavior, transient response, and small-signal stability. A mathematical model is developed in the synchronous reference frame to facilitate analytical understanding and controller design. The effectiveness of the proposed approach is validated through time-domain simulations under various load disturbances, with particular emphasis on inductive loading conditions.

II. SYSTEM DESCRIPTION AND MODELING

A. System Configuration

This paper considers a three-phase grid-forming inverter supplying a local load through an output filter, as shown in Fig. 1. An LC filter is employed at the inverter output to attenuate switching harmonics and shape the output voltage waveform. Since the load operates in the megawatt range, the inclusion of a step-up transformer is required. The transmission line is subsequently modeled as a series RL circuit. Finally, the load is connected at the receiving end.

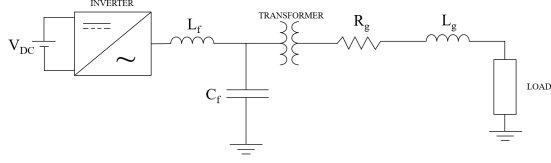


Fig. 1. Configuration of the grid-forming inverter

The DC-side dynamics are neglected in this study, assuming a sufficiently stiff DC-link voltage provided by an upstream energy source. This assumption allows the analysis to focus on the AC-side dynamics and control interactions relevant to grid-forming operation.

B. Reference Frame

To facilitate control design and dynamic analysis, the system equations are expressed in a synchronous rotating reference frame (dq-frame) that rotates at an angular frequency ω . The three-phase abc variables are first transformed into the stationary $\alpha\beta$ reference frame and then into the synchronous rotating dq -frame via Clark and Park transformation. This allows balanced sinusoidal quantities to appear as DC values in steady state, thereby simplifying control design and dynamic analysis [7]. In the synchronous dq reference frame, balanced three-phase AC quantities become approximately constant in steady-state operation. As a result, conventional linear PI controllers can be effectively applied for voltage and current regulation without the steady-state tracking error that would arise in the stationary abc frame. This transformation also simplifies stability analysis because the system dynamics can be represented using time-invariant differential equations.

Clarke Transformation:

$$\begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix} \begin{bmatrix} x_a \\ x_b \\ x_c \end{bmatrix} \quad (1)$$

Park Transformation:

$$\begin{bmatrix} x_d \\ x_q \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_\alpha \\ x_\beta \end{bmatrix} \quad (2)$$

Angle-Frequency Relation:

$$\dot{\theta} = \omega \quad (3)$$

C. LC Filter

In a grid-forming (GFM) inverter, an LC filter is used at the output stage to suppress the high-frequency harmonics produced by PWM switching and to ensure a near-sinusoidal voltage [3] at the point of common coupling. Since a GFM inverter operates as a controlled voltage source, the LC filter is not merely a harmonic mitigation element but an integral

part of the voltage-forming mechanism. The filter inductor limits current ripple and reduces switching stress, while the filter capacitor smooths the output voltage and provides a local energy buffer.

D. Load and Line Model

In microgrids, short transmission lines are typically modeled using series resistance R_{line} and inductance L_{line} . Rather than specifying the internal resistance R_{tr} and inductance L_{tr} of the transformer separately, these parameters are combined with the line impedance. Since normal loads often include rotating machines, which are generally inductive, a balanced three-phase inductive load is employed for modeling purposes.

III. DROOP-ONLY GRID-FORMING CONTROL

A. Droop Control Concept

Droop control is a widely adopted decentralized control strategy for grid-forming inverters. It enables the inverter to regulate its output frequency and voltage magnitude based on active and reactive power, thereby emulating the behavior of conventional synchronous generators. The droop relations are defined as:

$$\omega = \omega^* - k_p (P - P^*) \quad (4)$$

$$V = V^* - k_q (Q - Q^*) \quad (5)$$

where ω is the inverter output angular frequency and V is the voltage magnitude. The variables P and Q denote the measured active and reactive power at the PCC, respectively. The parameters ω^* and V^* represent the nominal frequency and voltage setpoints, while P^* and Q^* are the active and reactive power references. The coefficients k_p and k_q correspond to the active and reactive power droop gains. Physically, the active-power droop equation emulates the frequency response characteristic of a synchronous generator, while the reactive-power droop equation regulates voltage magnitude according to reactive power demand. Consequently, the inverter can autonomously adjust its operating point following load variations without requiring centralized communication.

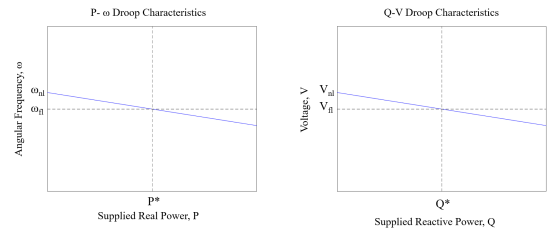


Fig. 2. P - ω and Q - V droop

The droop gains determine the sensitivity of the inverter frequency and voltage to power deviations. Higher droop gains enhance power-sharing capability among multiple inverters; however, excessively large gains may reduce system damping and compromise stability margins.

B. Power calculation

In the dq -frame, the measured PCC voltages and currents are used to calculate the instantaneous active and reactive power for per unit system:

$$P = v_d i_d + v_q i_q \quad (6)$$

$$Q = v_q i_d - v_d i_q \quad (7)$$

In the dq -frame, if the d -axis is aligned with the PCC voltage, we have:

$$v_q = 0 \quad (8)$$

The instantaneous active and reactive power then simplify to:

$$P = v_d i_d \quad (9)$$

$$Q = -v_d i_q \quad (10)$$

These simplified expressions illustrate the decoupled relationship between active power and d -axis current, and between reactive power and q -axis current. This property is particularly useful for independent active and reactive power regulation in grid-forming inverter control.

C. Characteristics and Limitations

Droop-only control enables primary load sharing and supports islanded operation without requiring communication among inverters. Despite these advantages, it exhibits several inherent limitations. High droop gains increase frequency sensitivity and can reduce stability margins if not carefully tuned. Limited inherent damping may lead to lightly damped oscillations following large load disturbances. Furthermore, the lack of inner current control or explicit current limiting can result in over-current during faults or severe disturbances, increasing the risk of protection trips and device stress. Reactive power disturbances further worsen the dynamic response by inducing oscillations (Figure 3). These limitations motivate the incremental introduction of inner voltage and current control loops, discussed in Section IV.

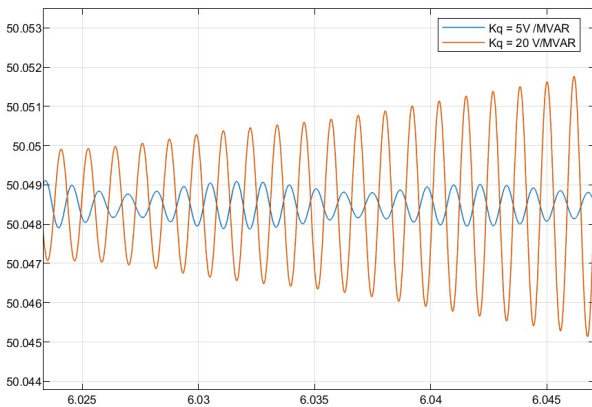


Fig. 3. Effect of load disturbance on frequency

IV. PROGRESSIVE DEVELOPMENT OF INNER CONTROL LOOPS

A. Power measurement filter

In droop-controlled grid-forming inverters, the instantaneous active and reactive power at the PCC, P and Q , are used as inputs to the droop control equations. However, these signals contain high-frequency components due to inverter switching and grid dynamics. To ensure stable operation, P and Q are passed through a first-order low-pass filter before being applied to the droop controller [2]. The filter is represented in the s -domain as

$$H(s) = \frac{1}{T_p s + 1}, \quad (11)$$

Without filtering, switching harmonics and measurement noise may propagate into the droop controller and excite oscillatory modes. Therefore, the power measurement filter acts as an important damping-support element that improves the robustness of the outer droop loop. This filtering smooths the power signals, reduces high-frequency noise, and prevents excitation of lightly damped modes, while still allowing the droop controller to respond to primary frequency and voltage deviations in a timely manner.

B. Inner Voltage Control Loop

Although droop control determines the voltage reference, it does not directly regulate the PCC voltage dynamics. Therefore, an inner voltage control loop is introduced to ensure accurate voltage tracking and faster transient response during load disturbances. The inner voltage control loop regulates the point of common coupling (PCC) voltage in the synchronous dq reference frame using a proportional–integral (PI) controller. The voltage tracking error is defined as

$$\mathbf{e}_v = \mathbf{v}_{\text{pcc},dq}^* - \mathbf{v}_{\text{pcc},dq}, \quad (12)$$

where $\mathbf{v}_{\text{pcc},dq}^*$ and $\mathbf{v}_{\text{pcc},dq}$ denote the reference and measured PCC voltages, respectively.

Based on the voltage error, the PI controller generates the reference current for the inner current control loop as

$$\mathbf{i}_{f,dq}^* = K_{pv} \mathbf{e}_v + K_{iv} \int \mathbf{e}_v dt, \quad (13)$$

where K_{pv} and K_{iv} are the proportional and integral gains of the voltage controller.

In the synchronous reference frame, the PCC voltage dynamics are inherently coupled due to the rotating coordinate transformation. Considering the filter capacitor C_f , the voltage dynamics can be expressed as

$$C_f \frac{dv_{\text{pcc},d}}{dt} = i_{f,d} - i_{\text{load},d} + \omega C_f v_{\text{pcc},q}, \quad (14)$$

$$C_f \frac{dv_{\text{pcc},q}}{dt} = i_{f,q} - i_{\text{load},q} - \omega C_f v_{\text{pcc},d}, \quad (15)$$

where ω is the angular frequency of the synchronous reference frame and $i_{\text{load},dq}$ denotes the load current. The terms $\pm \omega C_f v_{\text{pcc},q/d}$ represent the cross-coupling between the d - and q -axis voltage components.

where the voltage cross-coupling compensation term is defined as

$$\mathbf{i}_{cc,dq} = \begin{bmatrix} -\omega C_f v_{pcc,q} \\ \omega C_f v_{pcc,d} \end{bmatrix}. \quad (17)$$

C. Inner Current Control Loop

$$\mathbf{e}_i = \mathbf{i}_{f,dq}^* - \mathbf{i}_{f,dq}, \quad (18)$$

The inverter voltage reference in the synchronous dq frame is generated as

where K_{pc} and K_{ic} are the proportional and integral current control gains, respectively, and $\mathbf{v}_{cc,dq}$ represents the feedforward cross-coupling compensation term:

which compensates for the coupling effects between the d - and q -axis currents in the synchronous reference frame.

To protect the inverter against over-current conditions during large disturbances, load transients, or short-circuit events, a current limiting mechanism is incorporated into the inner current control loop [1]. The magnitude of the reference current vector is constrained as

where $I_{\max}$ denotes the maximum allowable inverter current.

$$\mathbf{i}_{f,dq}^{*,\text{lim}} = \begin{cases} \mathbf{i}_{f,dq}^* & \text{if } \|\mathbf{i}_{f,dq}^*\| \leq I_{\max}, \\ \frac{I_{\max}}{\|\mathbf{i}_{f,dq}^*\|} \mathbf{i}_{f,dq}^* & \text{otherwise} \end{cases} \quad (22)$$

E. Virtual Impedance

$$\begin{bmatrix} v_{v,d} \\ v_{v,q} \end{bmatrix} = \begin{bmatrix} R_v & -\omega L_v \\ \omega L_v & R_v \end{bmatrix} \begin{bmatrix} i_d \\ i_q \end{bmatrix} \quad (23)$$

F. Overall Control Structure

[illegible]

Fig. 4. Overall control structure of the proposed single grid-forming inverter.

V. SMALL-SIGNAL ANALYSIS OF THE REDUCED-ORDER MODEL

A. Droop-Only Reduced-Order Model

The GFM operates with conventional active and reactive power droop control,

$$\omega = \omega_0 - k_p(P_f - P^*), \quad (24)$$

$$V = V_0 - k_q(Q_f - Q^*), \quad (25)$$

where k_p and k_q denote the active power–frequency and reactive power–voltage droop coefficients, respectively.

The measured active and reactive powers are obtained through first-order low-pass filters,

$$\dot{P}_f = -\omega_c P_f + \omega_c P, \quad (26)$$

$$\dot{Q}_f = -\omega_c Q_f + \omega_c Q, \quad (27)$$

where ω_c is the filter bandwidth.

Assuming a predominantly inductive grid connection and small phase-angle deviations, the injected active and reactive powers can be approximated by

$$P \approx \frac{V_g V_0}{X_g} \delta, \quad Q \approx \frac{V_0}{X_g} (V - V_g), \quad (28)$$

where X_g denotes the grid reactance and δ is the inverter angle with respect to the grid.

Linearizing around an equilibrium point and defining the state vector

$$x = [\Delta\delta \quad \Delta P_f \quad \Delta Q_f \quad \Delta V]^\top, \quad (29)$$

the droop-only small-signal dynamics can be written as

$$\dot{x} = A_{\text{droop}} x, \quad (30)$$

The eigenvalues of A_{droop} characterize the intrinsic synchronization and voltage dynamics of the droop-controlled GFM [5].

B. Reduced-Order Model with Virtual Damping

To represent the stabilizing effect of inner control loops without increasing model complexity, the inner control action is modeled as a virtual damping term acting on the angle dynamics. This approach reflects the damping contribution of fast inner voltage and current controllers to power oscillations.

The modified angle dynamics after linearization are given by

$$\dot{\delta} = -k_p P_f - D_v \delta, \quad (31)$$

where D_v denotes the virtual damping coefficient introduced by the inner control loop.

Compared to the droop-only case, the virtual damping term introduces an additional negative real component in the angle-related eigenvalues, shifting system modes further into the left half-plane. This demonstrates that inner control loops can significantly enhance small-signal stability even when represented by minimal-order dynamics.

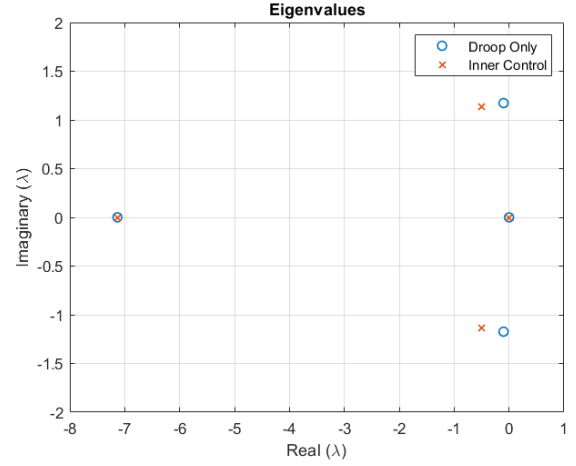


Fig. 5. Effect of damping on eigenvalues

VI. RESULTS AND DISCUSSION

The objective of this section is to evaluate the dynamic behavior and stability characteristics as the control architecture progresses from droop-only control to inner-loop-augmented and advanced control structures. The system parameters used for simulation are summarized in Appendix A (Table I).

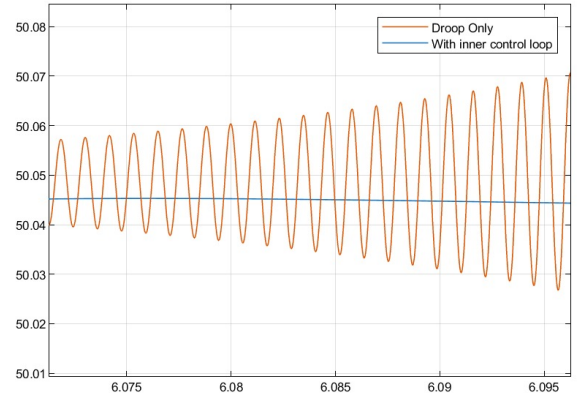


Fig. 6. Effect of sudden load rejection on frequency

Figure 6 shows the inverter frequency response following a sudden load rejection. Under droop-only operation, the system exhibits poorly damped oscillations with a peak-to-peak frequency variation more than 0.04 Hz. Moreover, the oscillation amplitude gradually increases with time, indicating insufficient damping and reduced stability margin. In contrast, with inner voltage and current control loops, the oscillation magnitude is reduced to approximately 0.001 Hz, corresponding to more than 95% attenuation.

A similar effect is observed in Fig. 7. In the droop-only case, the inverter exhibits sustained low-frequency oscillations with a frequency deviation between approximately 49.955 Hz and 50.055 Hz, corresponding to nearly 0.1 Hz peak-to-peak oscillation. These oscillations persist throughout the interval,

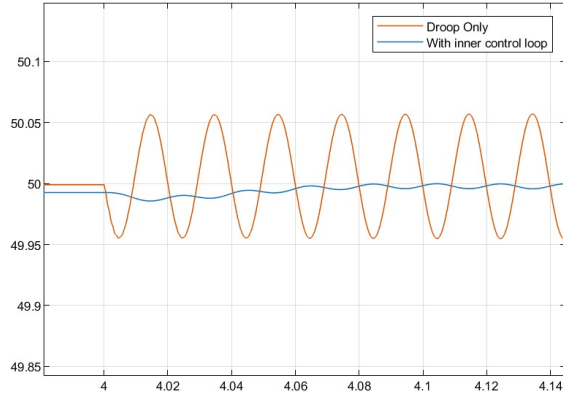


Fig. 7. Impact of purely inductive load disturbance

confirming weak damping under highly inductive conditions. With inner control loops, the oscillation amplitude reduces to within approximately ± 0.005 Hz, representing nearly a 90% reduction.

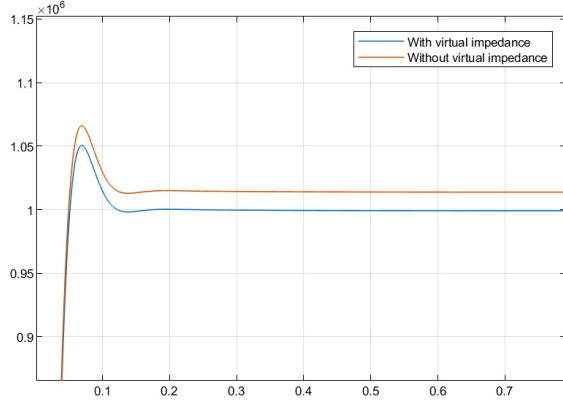


Fig. 8. Precise power control by virtual impedance

Figure 8 compares the active power response of the grid-forming inverter with and without virtual impedance. With virtual impedance, the peak overshoot reduces from approximately 1.065 MW to 1.05 MW, the response converges faster and provides precise power output control (Generating exactly 1 MW real power). These results confirm that virtual impedance reshapes the inverter output impedance, enhancing power regulation and reducing transient oscillations in inductive networks.

VII. CONCLUSION

This paper presented a structured and progressive design methodology for grid-forming inverter control, starting from conventional droop-only operation and extending to advanced inner-loop-based control structures. The study demonstrated how successive control layers influence voltage regulation, damping behavior, and small-signal stability characteristics. Rather than introducing an entirely new control algorithm, the

contribution of this work lies in providing a unified analytical and design-oriented framework that clearly illustrates the dynamic impact of each control enhancement step.

APPENDIX

TABLE I
GRID-FORMING INVERTER PARAMETERS

Symbol	Parameter	Value
S_{base}	Rated apparent power	2 MVA
V_{base}	Rated line-line voltage	400 V
f_{base}	Rated frequency	50 Hz
V_{dc}	DC-link voltage	1000 V
k_p	Active power–frequency droop gain	0.5 Hz/MW
k_q	Reactive power–voltage droop gain	20 V/MVAR
f_c	Power measurement filter cutoff frequency	10 Hz
K_{pv}	Voltage controller proportional gain	5
K_{iv}	Voltage controller integral gain	10
K_{pc}	Current controller proportional gain	0.2
K_{ic}	Current controller integral gain	1
L_f	Filter inductance	8 μ H
C_f	Filter capacitance	6 mF
R_d	Filter internal resistance	0.1 m Ω
R_g	Grid side resistance	0.75 Ω
L_g	Grid side inductance	5 mH
n_t	Transformer turns ratio	400 V / 33 kV
P	Operating active power	0–2 MW
Q	Operating reactive power	0–1.2 MVAR
pf	Power factor	0.8 lag–lead

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